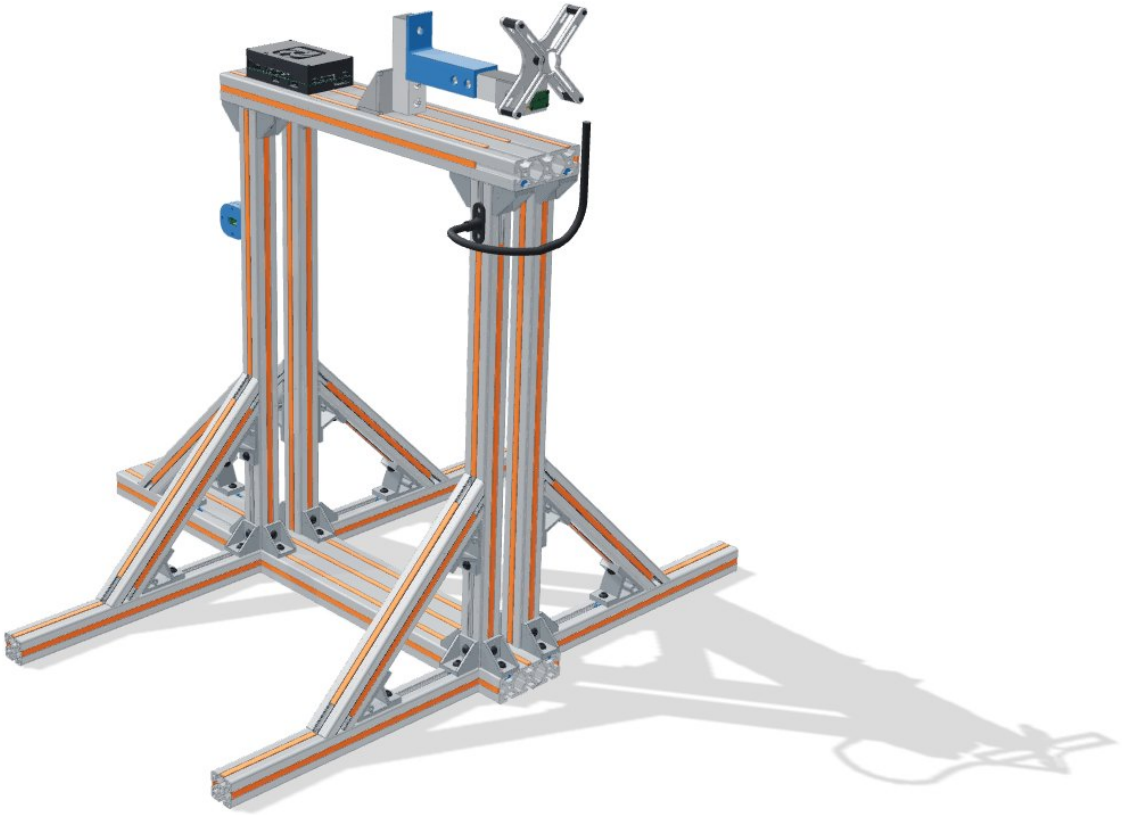
DynoX thrust stand

RTS Ultra 50

User manual · 50 kgf thrust and torque test stand



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Designed and manufactured by REUDE Technologies Private Limited, Chennai, India. Read the safety instructions in Section 3 before assembling or operating this equipment.

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1. About this manual

This manual explains how to assemble, wire, calibrate, operate and maintain the RTS Ultra 50. It is written for technicians, engineers and lab staff with basic mechanical and electrical skills. Keep it with the stand and pass it on with the equipment.

How to read the safety notes

WARNING A hazard that can cause serious injury. Follow the instruction exactly.

CAUTION A risk of damage to the stand, the sensors or the measurement.

NOTE Useful information that makes the job easier or the result better.

Interactive 3D model

Every component in this manual is numbered the same way as in the DynoX 3D viewer. Open the viewer to rotate the stand, see the exploded sub-assemblies, and lay every component out on the table.

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2. Product overview

The RTS Ultra 50 is a heavy-duty thrust stand for large drone motors up to 50 kgf of thrust. A four-pillar box tower stands on a 900 mm base rail with four side outriggers and six diagonal braces, giving a footprint about 1 m wide. The sensing head uses two 100 kg load cells and stainless-steel brackets, with an adjustable X-pattern motor plate and a gooseneck RPM sensor holder. The ROTRIX DAQ unit sits on the top beam beside the head and streams all readings to the ROTRIX software.

Specifications

Item	Specification
Rated thrust	0 to 50 kgf
Measured parameters	Thrust, torque, RPM, voltage, current, vibration and temperature
Force sensing	Two 100 kg beam load cells, one for thrust and one for torque (2× overload margin)
Motor interface	X-pattern 5 mm stainless plate, 125 × 125 mm, slotted adjustable arms, on 20 mm M6 spacers
Speed sensing	Optical RPM sensor on a flexible gooseneck holder
Electrical sensing	Inline current sensor module on the rear pillar; voltage measured by the DAQ
Data acquisition	ROTRIX DAQ unit on the top beam; USB link
Software	ROTRIX test control and logging; ROTRIDASH analytics (optional)
Frame	40-series aluminium extrusions with orange slot covers (REUDE standard)
Footprint	About 1,000 × 900 mm
Overall height	About 960 mm
Thrust axis height	About 900 mm above the floor

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Main components

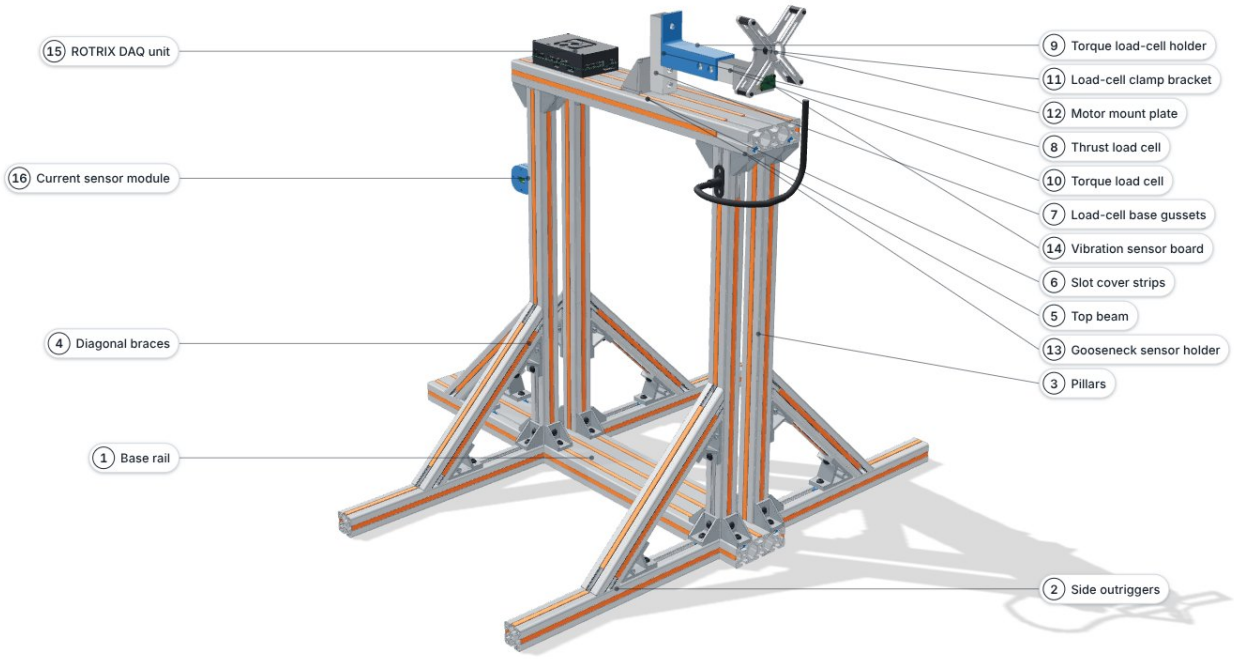


Figure 2.1 RTS Ultra 50 assembled, with the main components numbered.

Exploded view

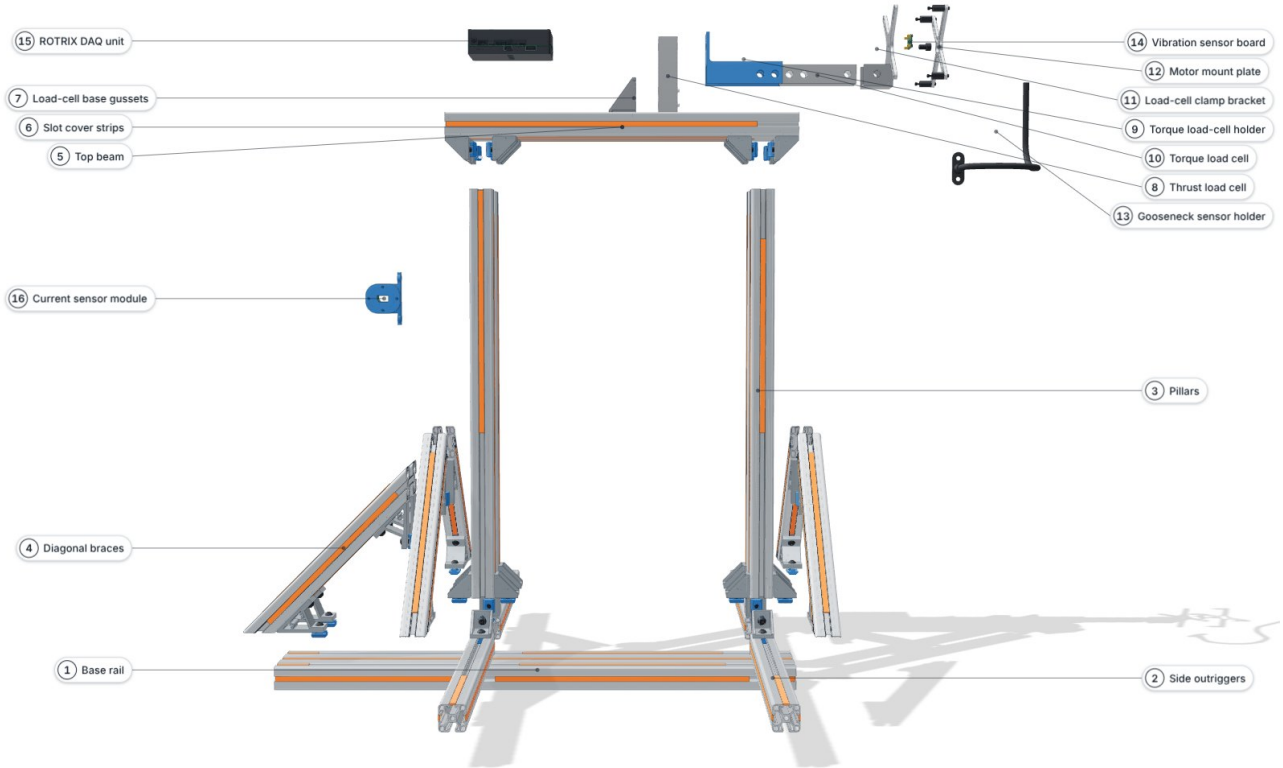


Figure 2.2 Exploded side view. Each sub-assembly moves as one block, in the order it is assembled.

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3. Safety instructions

WARNING A spinning propeller can cause severe injury. Never approach the stand while a battery is connected, and keep everyone out of the propeller plane during a test.

Assembly team

At least two people are required, and three when lifting the top beam onto the pillars.

People and protection

- Read this manual completely before assembling or operating the stand.
- Wear safety glasses and gloves during assembly. Wear safety glasses and hearing protection during every test run.
- Tie back long hair and remove loose clothing, lanyards and jewellery before approaching a running motor.
- Never reach over, under or past a propeller while a battery is connected, even if the motor is stopped.

Test area

- Place the stand on a level, rigid floor or bench and anchor it with the supplied clamps before any powered test.
- Mark an exclusion zone around the stand. Nobody may stand in the plane of the propeller disc while a battery is connected.
- Keep the area clear of loose paper, tools, cable offcuts and anything that the propeller wash can lift.
- Test indoors or under cover. The sensors and the DAQ unit are not weatherproof: keep them away from rain, condensation and dust.

Electrical and battery

- Connect the battery last and disconnect it first. Keep the throttle at zero and the ESC disarmed while connecting.
- Use cables, connectors and fuses rated above the maximum test current. Check insulation before every session.
- Fit an emergency-stop or kill switch in the power line, within reach of the operator, and test it before each session.
- Charge, store and transport lithium batteries in fire-safe bags or containers, and never leave a charging battery unattended.
- When the operator works at a distance, only the digital data link and the emergency-stop loop should run to the operator station. Do not extend the analogue sensor cables.

Operation

- Never exceed the rated thrust of the stand. Stop the test if a reading approaches the limit.
- Before each run, check that every bolt is tight, the propeller is undamaged and balanced, and its hub bolts are secure.
- Start every new motor and propeller combination at low throttle and increase in steps while watching vibration and temperature.
- Stop immediately on unusual noise, rising vibration, smoke, or any reading that does not make sense.

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- Do not lean on, lift or carry the stand by the sensing head. Loads on the head can permanently damage the load cells.

CAUTION The rated thrust of this stand is 50 kgf. Loads above the rating can permanently damage the load cells and void the calibration.

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4. Package contents

Check every item against this list before you start. If anything is missing or damaged, contact REUDE support before assembly. Do not start with incomplete or faulty parts.

No.	Component	Code	Size	Material	Qty
1	**Base rail**	40 × 120 aluminium extrusion	40 × 120 mm section, 900 mm long	Anodised aluminium extrusion	1
2	**Side outriggers**	40 × 40 profile	40 × 40 × 440 mm	Anodised aluminium extrusion	4
3	**Pillars**	40 × 40 profile	40 × 40 × 700 mm	Anodised aluminium extrusion	4
4	**Diagonal braces**	40 × 40 profile at 45°	40 × 40 mm profile, about 400 mm long	Anodised aluminium extrusion	6
5	**Top beam**	40 × 120 aluminium extrusion	40 × 120 mm section, 600 mm long	Anodised aluminium extrusion	1
6	**Slot cover strips**	Orange T-slot covers · REUDE standard	9 mm wide, cut to length for each exposed slot	Flexible PVC, REUDE orange	1 set
7	**Load-cell base gussets**	5 mm stainless steel	80 × 60 × 45 mm	5 mm stainless steel	4
8	**Thrust load cell**	100 kg beam load cell (vertical)	32 × 32 × 130 mm	Precision strain-gauge load cell, 100 kg capacity	1
9	**Torque load-cell holder**	Coupling block	42 × 87 × 130 mm	Machined block	1
10	**Torque load cell**	100 kg beam load cell (horizontal)	32 × 32 × 130 mm	Precision strain-gauge load cell, 100 kg capacity	1
11	**Load-cell clamp bracket**	5 mm stainless steel	125 × 132 × 50 mm	5 mm stainless steel	1
12	**Motor mount plate**	X-pattern, adjustable arms	125 × 125 mm, 5 mm thick, on 20 mm spacers	5 mm stainless steel, steel spacers and screws	1
13	**Gooseneck sensor holder**	Flexible sensor arm	About 138 × 179 × 144 mm	Flexible steel gooseneck with clamp	1
14	**Vibration sensor board**	Accelerometer	About 32 × 22 mm on 6 mm spacers	Circuit board, brass spacers	1
15	**ROTRIX DAQ unit**	STM32-based acquisition	86 × 140 × 47 mm	Custom PCB with signal conditioning, in a 3D-printed enclosure	1
16	**Current sensor module**	Inline current sensor	58 × 90 × 27 mm case	3D-printed case, current sensor board	1

Brackets and fixings

No.	Fixing	Code	Size	Material	Qty
17	**90° corner brackets**	40 × 40 bracket	40 × 40 × 35 mm	Die-cast aluminium	21

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No.	Fixing	Code	Size	Material	Qty
18	**45° brace brackets**	Diagonal end brackets	35 × 58 × 80 mm	Die-cast aluminium	12
19	**T-slot nuts**	Series 8 tee nuts	Series 8, for 40 mm profiles	Printed polymer	64
20	**M8 button-head screws**	Bracket screws	M8	Stainless steel	64

Component importance

Critical parts are in the load path or the measurement chain. **Important** parts set the stiffness and alignment of the frame. **Standard** parts are for convenience and protection.

No.	Component	Importance	Why it matters
1	Base rail	**Critical**	Up to 50 kgf of thrust acts high above the floor. The long rail turns that tipping force into a manageable load on the floor.
2	Side outriggers	**Critical**	Motor torque tries to roll the stand sideways. A 1 m wide footprint keeps it planted at full power.
3	Pillars	**Critical**	Four pillars share the 50 kgf load and resist twisting from motor torque far better than a single mast.
4	Diagonal braces	**Critical**	They form triangles with the base and pillars, so the tower cannot lean sideways or fore-aft when the motor pushes or pulls.
5	Top beam	**Critical**	It is the platform for the load cells. Any flex here reads as false thrust, so it must be level and fully bolted.
6	Slot cover strips	**Standard**	They keep dust, swarf and small parts out of the slots, cover the sharp slot edges, hold cables neatly, and give the stand its signature orange accent.
7	Load-cell base gussets	**Critical**	They are the fixed end of the thrust measurement. A rigid anchor means the load cell sees all of the thrust.
8	Thrust load cell	**Critical**	A 100 kg cell on a 50 kgf stand gives a 2× overload margin while keeping high resolution.
9	Torque load-cell holder	**Important**	It keeps the two sensors at right angles so each one reads only its own force.
10	Torque load cell	**Critical**	Torque is needed for mechanical power and propeller efficiency figures.
11	Load-cell clamp bracket	**Critical**	It carries every newton of thrust and torque from the motor into the sensors, so it must not flex.
12	Motor mount plate	**Critical**	The slotted arms let you centre almost any heavy-lift motor exactly on the thrust axis.
13	Gooseneck sensor holder	**Important**	It lets you aim the RPM sensor at the blade path for any propeller size.
14	Vibration sensor board	**Important**	Mounted right behind the motor, it picks up imbalance and resonance early.
15	ROTRIX DAQ unit	**Critical**	It samples thrust, torque, RPM, voltage, current, vibration and temperature in sync and sends them to your PC over USB.
16	Current sensor module	**Critical**	Current is half of electrical power, so it is needed for every efficiency figure.

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All components laid out

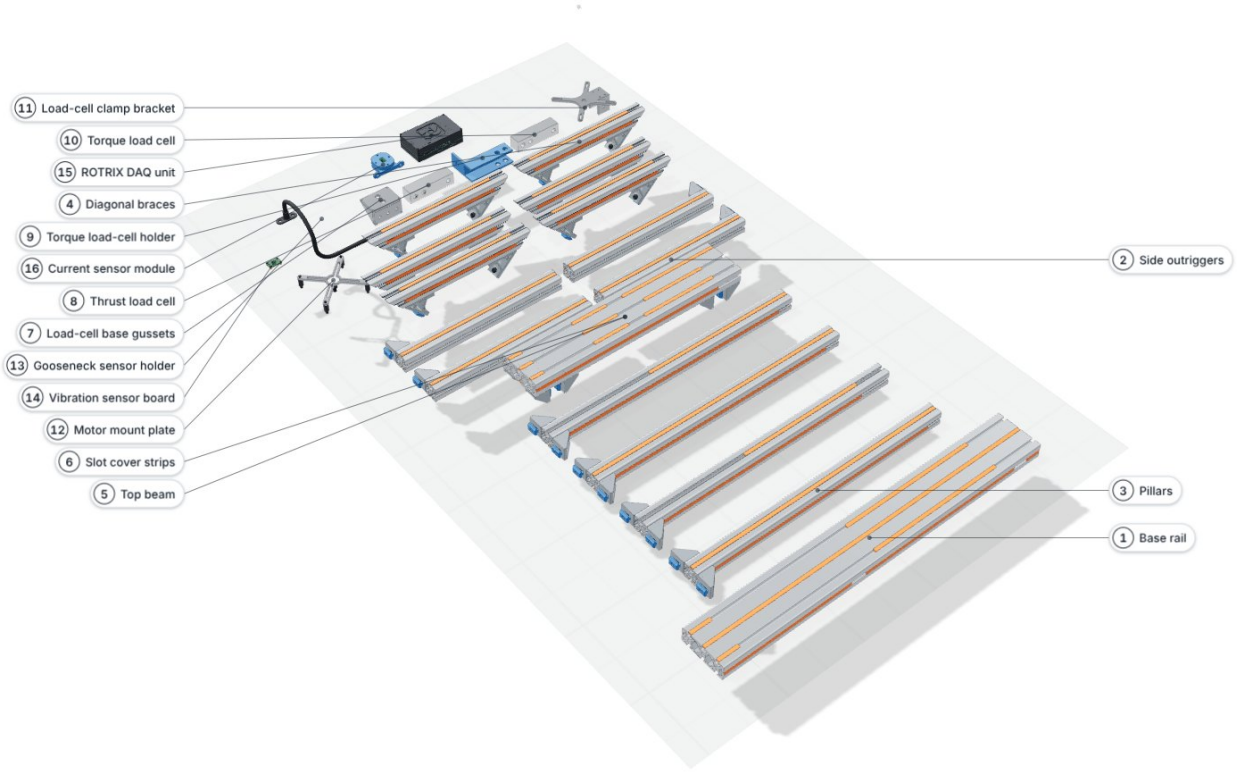


Figure 4.1 The components laid out on the table, as shipped.

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5. Tools and tightening torques

Tools you will need

- Hex keys 2.5, 5 and 6 mm
- Spanner or socket set
- Torque wrench
- Spirit level, try square and tape measure
- Cable ties, a side cutter and a sharp knife (for the slot covers)
- Multimeter and clamp meter
- Hand tachometer
- Calibration weights (for example 5, 10 and 20 kg) with cord and a heavy-duty pulley
- Masonry drill and M12 anchors for concrete floors

Recommended tightening torques

Fastener	Where used	Typical tightening
M3 screws	Printed parts, sensor boards, DAQ enclosure	Hand tight, about 0.5 to 1 Nm. Do not crack the printed parts.
M6 bolts	Motor to mounting plate, plate to bracket	About 8 Nm, with thread-locker on motor bolts
M8 button-head bolts	Brackets into T-slot nuts	About 12 to 15 Nm into steel T-nuts; snug only (about 4 Nm) into printed T-nuts. Tighten evenly around each joint.

NOTE Tighten in two passes: first snug every bolt in a joint, then bring each to the final torque. This keeps the frame square.

CAUTION Always load the T-slot nuts into a profile before a bracket, end cap or another profile closes that slot. A missing nut means taking the joint apart again.

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6. Assembly, step by step

Work through the steps in order. In each figure the parts added in that step are highlighted in orange; parts fitted in earlier steps are shown in their final position.

Step 1. Base rail

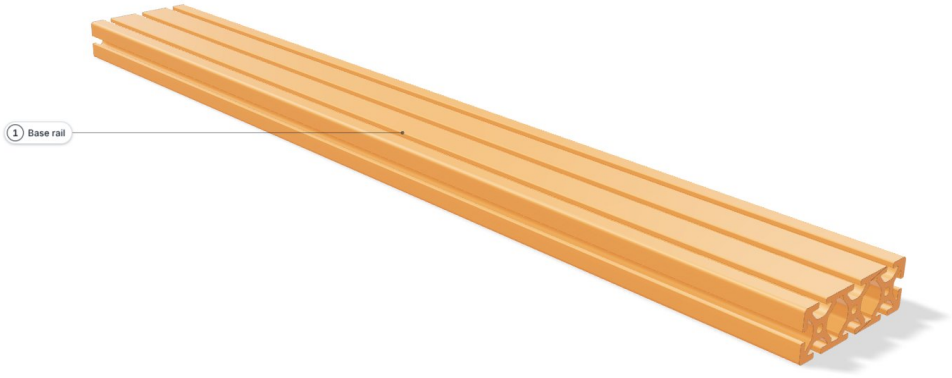


Figure 6.1 Base rail.

1. Load T-nuts into the 900 mm base rail at the four pillar positions and the two rear brace feet.
2. Place the rail on a level, rigid floor with the long end towards the rear.

CAUTION Load every T-nut the rail needs now. They cannot be added once brackets close the slots.

Step 2. Four pillars

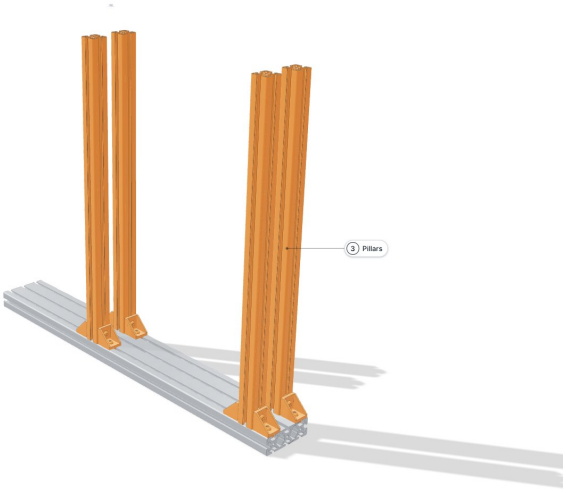


Figure 6.2 Four pillars.

1. Load T-nuts into each pillar for the top-beam brackets, the braces and the current sensor module.
2. Stand each 700 mm pillar on its position: two at the front and two at the rear.
3. Fix each pillar foot to the base rail with 90° corner brackets and M8 button-head screws.
4. Check every pillar is vertical in both directions, then tighten.

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Step 3. Side outriggers

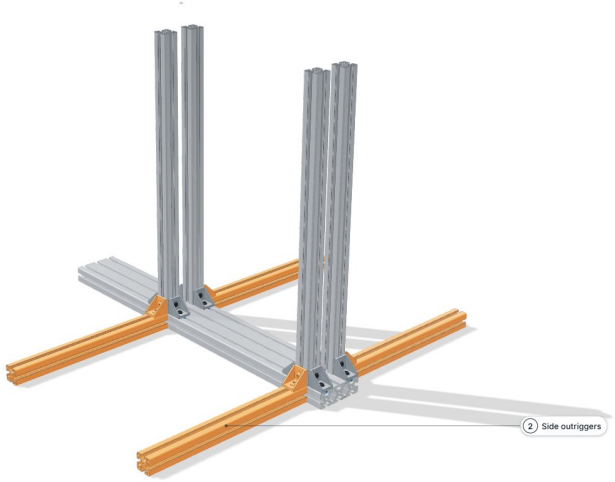


Figure 6.3 Side outriggers.

1. Load T-nuts into the four 440 mm outriggers for the corner brackets and side braces.
2. Bolt an outrigger to each side of the front and rear pillar feet with 90° corner brackets.
3. Check that all four outriggers sit flat on the floor and square to the base rail.

Step 4. Diagonal braces

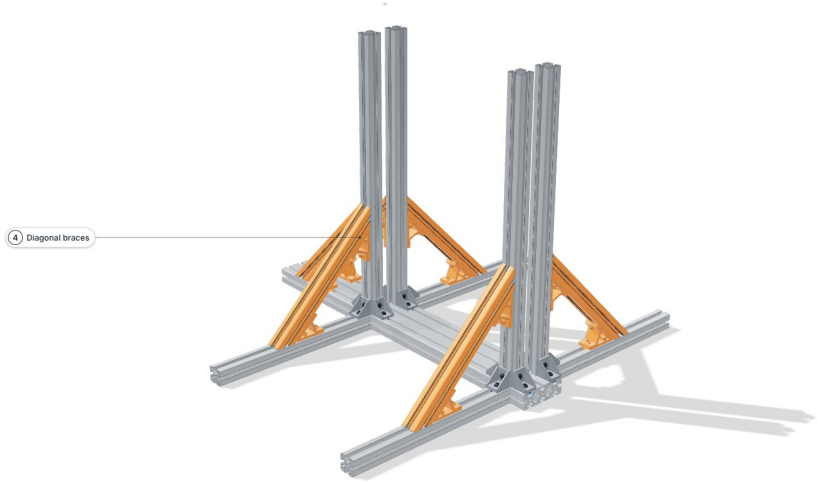


Figure 6.4 Diagonal braces.

1. Fit a 45° bracket at each end of all six braces.
2. Bolt the four side braces between the outriggers and the pillars.
3. Bolt the two rear braces between the rear of the base rail and the rear pillars.
4. Tighten evenly, then recheck that all four pillars are vertical.

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Step 5. Top beam



Figure 6.5 Top beam.

1. Load T-nuts into the top beam for the sensing head, the DAQ unit and the pillar brackets.
2. With two or three people, lower the 600 mm beam onto the four pillar tops, long overhang at the front.
3. Fix it to each pillar with 90° corner brackets and check that it is level along its length.
4. Anchor the base and outriggers to the floor (M12 anchors on concrete).

CAUTION Lift the top beam with enough people. The stand must be anchored before any powered test.

Step 6. Load-cell anchor gussets and thrust load cell

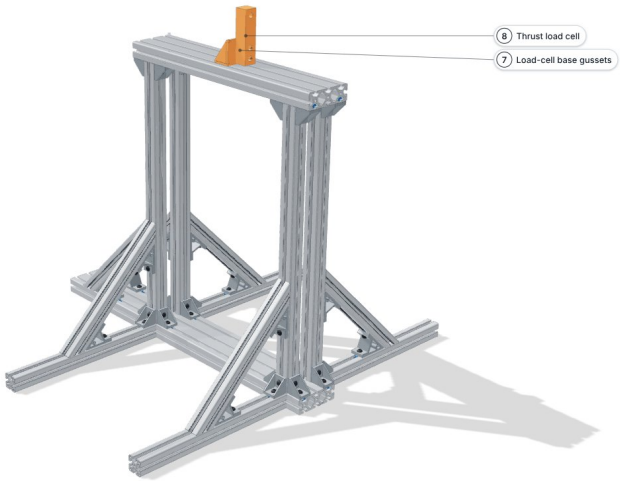


Figure 6.6 Load-cell anchor gussets and thrust load cell.

1. Bolt the four 5 mm stainless gussets to the top beam at the rear of the head position.
2. Bolt the 100 kg thrust load cell upright onto the gussets with its two cap screws.
3. Check that the arrow on the load cell points in the direction the thrust will push.

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Step 7. Torque holder, torque load cell and clamp bracket

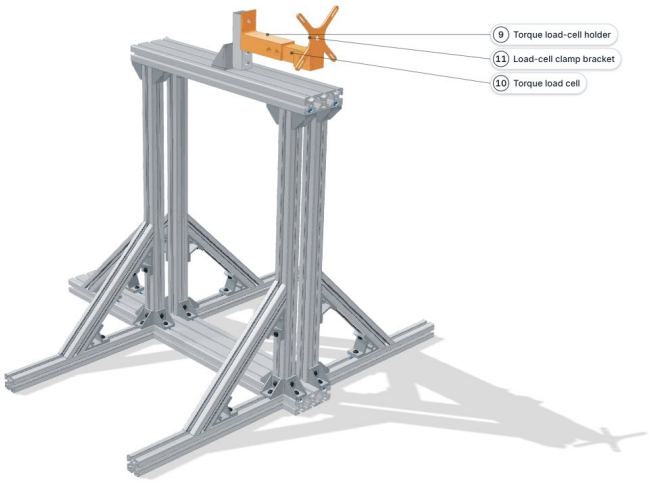


Figure 6.7 Torque holder, torque load cell and clamp bracket.

1. Bolt the torque load-cell holder to the free end of the thrust load cell.
2. Slide the 100 kg torque load cell into the holder and tighten its screws.
3. Fit the stainless clamp bracket on the front end of the torque load cell, square to the thrust axis.

CAUTION Never load the load cells by hand beyond the rated thrust, and never use the head as a lifting point.

Step 8. Motor mount plate, vibration board and gooseneck sensor holder

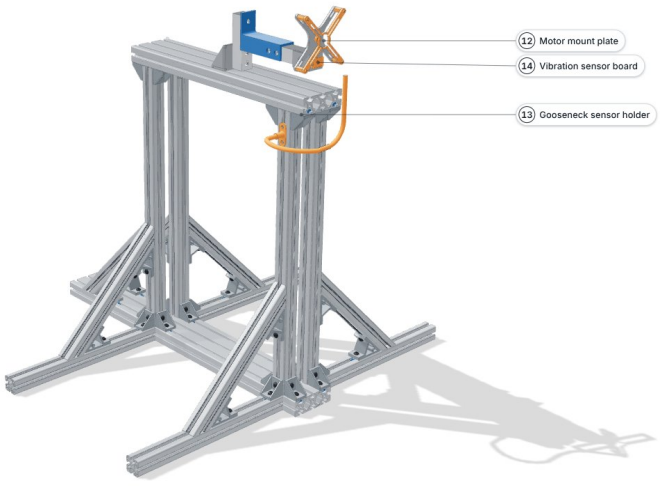


Figure 6.8 Motor mount plate, vibration board and gooseneck sensor holder.

1. Fix the X-pattern motor plate to the clamp bracket through the four 20 mm M6 spacers and the centre screw.
2. Fit the vibration board on its spacers behind the motor plate.
3. Clamp the gooseneck holder to the front of the top beam and fit the RPM sensor to it.
4. Slide the plate arms to match the motor bolt circle, then bolt the motor on, centred on the plate.

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Step 9. ROTRIX DAQ unit



Figure 6.9 ROTRIX DAQ unit.

1. Bolt the DAQ unit to the rear of the top beam through two T-nuts, with its connectors facing the sensing head.
2. Connect all sensor cables as described in Section 7.

CAUTION The DAQ unit is supplied assembled and sealed. Do not open the enclosure.

Step 10. Current sensor module



Figure 6.10 Current sensor module.

1. Fix the case base to the rear pillar.
2. Place the current sensor board, close the cover and screw it shut.
3. Run the battery-to-ESC lead through the sensor, with the arrow pointing from the battery to the ESC.

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Step 11. Orange slot cover strips (REUDE standard)

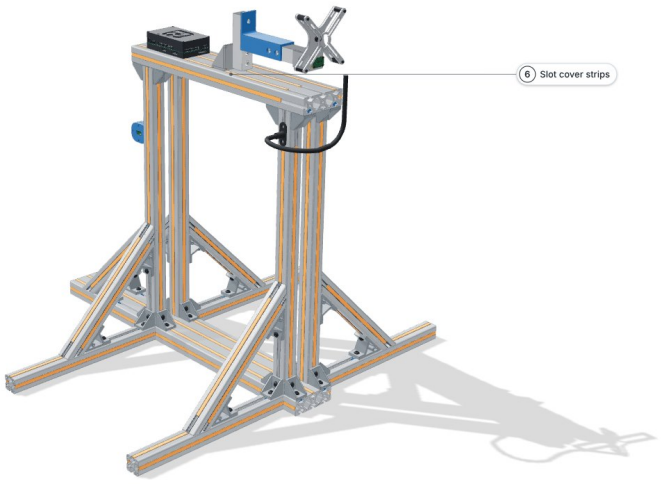


Figure 6.11 Orange slot cover strips (REUDE standard).

1. Check that every bolt in the frame is at its final torque and that all cables are routed.
2. Press an orange cover strip into each open T-slot, starting at one end and working along by hand.
3. Cut the strip flush at each bracket and at the profile ends with a sharp knife.
4. Where a cable runs inside a slot, lay it in the slot first and press the cover over it.

CAUTION Fit the covers last. They are easy to lift out later if you need to add a T-nut or re-route a cable.

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Sensing head, exploded

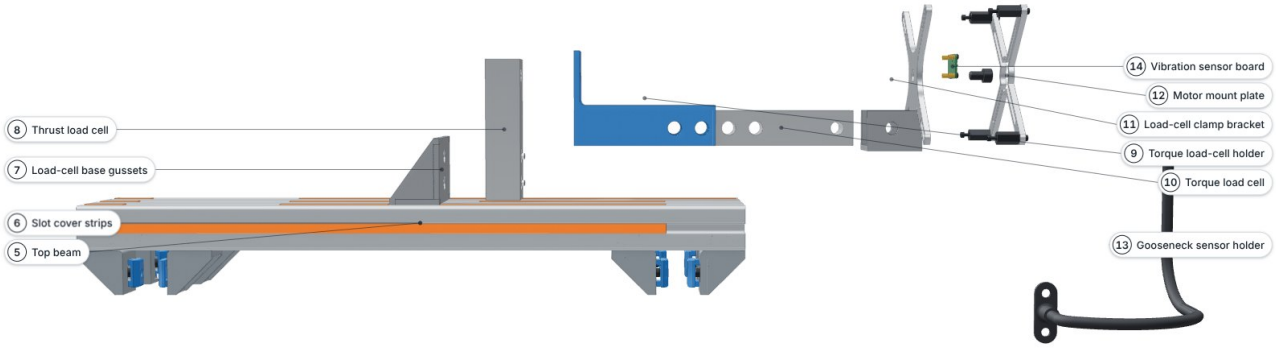


Figure 6.12 Exploded side view of the sensing head on the top beam.

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7. DAQ and electronics assembly

On the Ultra 50 the ROTRIX DAQ unit mounts on the rear of the top beam, so the load-cell cables stay short and away from the propeller. It is supplied as one sealed unit.

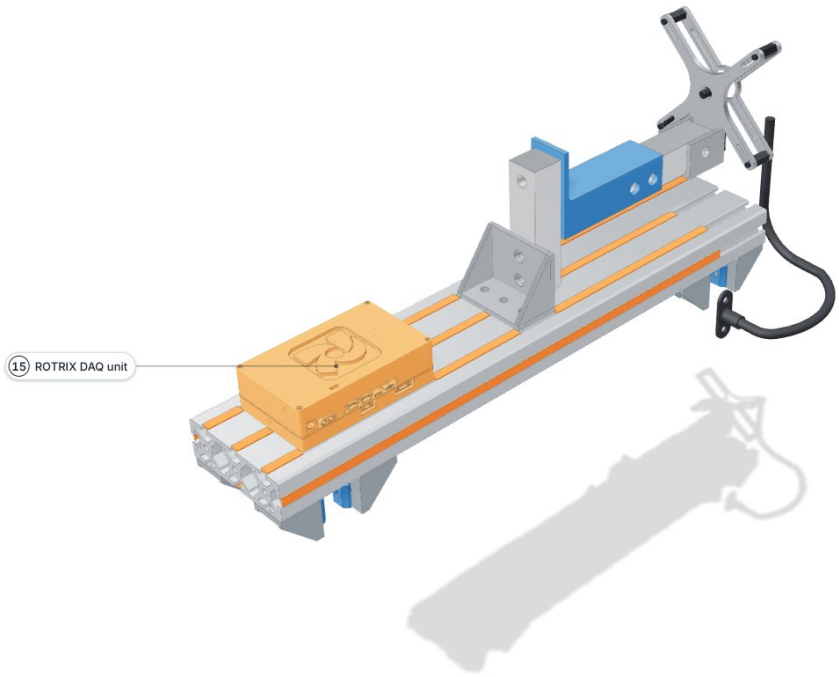


Figure 7.1 ROTRIX DAQ unit on the rear of the top beam, behind the sensing head.

Mounting and connecting

1. Slide two T-nuts into the rear slot of the top beam.
2. Bolt the DAQ unit to the beam with its connectors facing the sensing head.
3. Connect the thrust and torque load cells, the RPM sensor, the vibration board and the current sensor.
4. Connect the DAQ unit to the PC with the USB cable.

What the DAQ unit measures

- Thrust and torque from the load cells, through precision bridge amplifiers.
- Motor speed from the optical RPM sensor.
- Battery voltage and current in the power line to the ESC.
- Vibration and temperature from the sensor boards.
- All channels are time-stamped together and streamed to ROTRIX for live display, logging and reports.

CAUTION Connect and disconnect sensor cables only with the DAQ unit powered off and the battery disconnected.

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8. Wire routing to the sensors

Good cable routing matters as much as good mechanics. A cable that is pulled tight across the sensing head carries part of the load, and the load cells then measure the cable instead of the motor. Follow the routes and rules below.



Figure 8.1 Recommended cable routes. Signal cables and power leads run on separate sides of the frame.

	Cable
	Thrust load cell
	Torque load cells
	RPM sensor
	Vibration sensor
	Current sensor
	Battery power (through the current sensor)
	USB to PC

Routing table

Cable	From	Route	To
Thrust load cell	Upright 100 kg cell	Short service loop at the cell, along the top beam to the rear	DAQ thrust input
Torque load cell	Horizontal 100 kg cell	Service loop at the holder, along the top beam beside the thrust cable	DAQ torque input
RPM sensor	Gooseneck holder	Along the gooseneck, then in the top-beam slot under the slot cover	DAQ RPM input

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Cable	From	Route	To
Vibration sensor	Board behind the motor plate	Service loop, then along the top beam	DAQ vibration input
Current sensor signal	Module on the rear pillar	Up the rear pillar slot to the top beam	DAQ current input
Power leads	Battery at the rear	Through the current sensor, up the rear pillar, along the opposite side of the top beam	ESC, then motor
USB	DAQ unit	Down the other rear pillar and along the floor to the operator station	PC

Routing rules

1. **Leave a service loop.** Where a cable leaves the sensing head, form a loose loop about 50 mm across, so the cable carries no load when the head deflects.
2. **Tie to the fixed frame only.** Fix cables to the frame with ties about every 150 mm. Never tie a cable to the motor plate, the load cells or the free end of the sensing head.
3. **Keep signal and power apart.** Run sensor cables and power leads on opposite sides of the frame. Where they must cross, cross at a right angle.
4. **Stay out of the propeller disc.** Route every cable behind the motor plate and away from the plane of the propeller.
5. **Relieve strain at the DAQ.** Leave a little slack at each connector and label both ends of every cable.
6. **Screened load-cell cables.** Connect the screen at the DAQ end only.
7. **Remote operation.** If the operator sits away from the stand, extend only the USB data link and the emergency-stop loop. Never extend the analogue sensor cables.
8. **Current sensor direction.** Fit the current sensor with its arrow pointing from the battery towards the ESC.
9. **Use the slots.** On the fixed frame, lay cables inside the T-slots and press the orange slot covers over them. This keeps cables tidy, protected and clear of the propeller.

CAUTION After any change to the wiring, zero the load cells again and check that the zero reading stays steady for one minute before testing.

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9. Commissioning and calibration

Before first power-up

- All bolts tightened to the recommended torque, and the base anchored.
- Motor fitted and centred; propeller not yet fitted.
- All cables routed with service loops, and the emergency stop tested.

Connect and check the sensors

1. Connect the DAQ unit to the PC over USB and open ROTRIX.
2. Confirm that every sensor is detected and shows a sensible live value.
3. Allow 10 minutes for the load cells to warm up, then zero (tare) thrust and torque with the motor fitted and stationary.

Check thrust and torque with known loads

1. Thrust: tie a cord to the motor hub, run it over a low-friction pulley in line with the thrust axis, and hang known weights (5, 10 and 20 kg). Compare each reading with the weight.
2. Torque: fit a light lever arm of known length to the motor plate and hang a known weight from its end. The expected torque is weight × 9.81 × arm length (N·m, with the weight in kg and the length in m).
3. If any reading is outside the stated accuracy, run the calibration routine in ROTRIX and repeat the check.

Check RPM, voltage and current

- Compare the RPM reading with a hand tachometer at two or three throttle settings.
- Compare voltage and current with a multimeter and a clamp meter.

First run

1. Fit and balance the propeller, and clear the exclusion zone.
2. Connect the battery last. Arm the ESC from the operator station.
3. Run at about 10% throttle for 30 seconds, watching vibration, temperature and noise.
4. Increase in steps of 10% only while every reading stays stable.

10. Operating the stand

- Do not exceed 50 kgf of thrust. Set a thrust limit alarm in ROTRIX below the rating.
- Run a short warm-up before each test series and zero the load cells with the motor stopped.
- Use ROTRIX to run automated throttle steps and ramps, and ROTRIDASH to compare runs and produce PDF performance reports.
- Allow the motor and ESC to cool between long runs, and watch the temperature channels.
- After testing, disarm the ESC, disconnect the battery, then stop logging and save the data.

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11. Maintenance

When	Check
Before every session	Bolts on the sensing head and motor; propeller condition and balance; cable routing and service loops; clamps anchored; emergency stop works
Weekly in regular use	All frame bolts at the recommended torque; sensors clean; no cracks in brackets or printed parts
Every month	Zero and span check of the thrust and torque readings with known weights; RPM check against a hand tachometer
Every 6 months	Full calibration in ROTRIX; inspect cables and connectors; replace worn fasteners
After a crash or overload	Stop testing. Inspect the load cells and brackets, then run a full calibration before further use

NOTE Clean the sensors and the motor plate with a dry cloth after each session. Do not use solvents on the load cells or the circuit boards.

12. Troubleshooting

Symptom	Likely cause	What to do
Thrust reading drifts with the motor stopped	Load-cell cable under tension, or the load cells are still warming up	Re-route the cable with a loose service loop, wait 10 minutes after power-up, then zero again
Readings are noisy or jumpy	Loose bolts, an unbalanced propeller, or signal cables running beside power leads	Retighten the frame, balance the propeller, separate signal cables from the power leads
Torque reads a value with no load	Sensor block not square to the thrust axis, or the motor plate is touching the frame	Check alignment and clearances, then zero again
RPM shows zero or jumps	Sensor not aimed at the blade path, or too far away	Re-aim the sensor and move it closer, keeping it clear of the blade tips
DAQ unit not found by ROTRIX	USB cable, port or driver	Try another cable and port, check the driver, and restart ROTRIX
The stand creeps or rocks during a run	Clamps not anchored, or the floor is uneven	Stop, level the base and anchor every clamp

13. Support and contact

Contact	Details
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When contacting support, quote the document ID on the cover, the stand serial number, and the ROTRIX software version.

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